Sufjan Fana

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Education

University of California, Berkeley December 2025
Applied Mathematics, B.A. + Computer Science, B.A.
- Relevant Courses: Machine Learning and Data Analytics, Deep Neural Networks, Computer Vision and Computational Photography, User Interface and Design, Optimization Models in Engineering, Probability and Random Processes, Computer Architecture, Abstract Linear Algebra, Linguistic Science

Experience

Research Assistant: Linguistic Analysis August 2024 – June 2025
Alexandra Pfiffner, U.C. Berkeley
- Automated phonetic analysis in R, Python, Praat to segment Atchan and Nghlwa recordings
- Standardized ELAN and Praat workflows to produce consistent segmentation and labels used across the lab
- Analyzed implosive acoustics and delivered visuals that informed ongoing lab manuscripts

Data Informatics Intern August 2024 – December 2024
Museum of Vertebrate Zoology, U.C. Berkeley
- Built SQL queries and cleanup scripts that improved specimen searchability and curator workflows
- Beta-tested portal features, filed actionable bugs, and added Python, R, HTML scripts for data pulls
- Produced GIS layers and maps for amphibian records, later published with citations

Information Technology Intern January 2021 – August 2022
De Anza College / Foothill College
- Resolved high-volume tech tickets for 20,000 students and faculty, raising resolutions with clear guides
- Performed updates and software configuration across lab machines to improve performance and security
- Trained users on new tools and basic cyber awareness, wrote quick-start docs adopted by departments

Projects

Steering Analysis for Vision-Language-Action Models | *Python, PyTorch*
- Reproduced activation-level steering via token-space clustering on OpenVLA and validated behavior changes through robot rollouts; evaluated transfer across prompts and tasks
- Compared token projection against a fine-tuned sparse autoencoder probe on pi0.5's semantic bottleneck; analyzed latent geometry and found instruction semantics entangled with sensorimotor state
- Ran ablations on normalization and training mixtures using $\sim$10M instruction-token residual activations; documented failure cases and stability limits under distribution shift

Automatic Panorama Stitching and Seam Blending | *Python, NumPy, OpenCV*
- Estimated homographies with DLT + RANSAC using Harris corners, ANMS selection, MOPS-style 8$\times$8 descriptors, and Lowe ratio matching with a mutual check
- Warped images with inverse mapping and bilinear sampling, then blended overlaps with distance-transform alpha masks and least-squares gain compensation to hide exposure seams
- Built a photo mosaic system with color matching and patch assignment; compared matching strategies and tuned tile sizes for sharper reconstructions

Diffusion Image Generation and Editing | *Python, PyTorch, DeepFloyd IF*
- Implemented forward noising and iterative denoising with classifier-free guidance; reproduced SDEdit-style edits, RePaint-style inpainting, and hybrid/anagram image effects
- Built and trained a U-Net from scratch for MNIST generation and denoising; added sinusoidal time conditioning and class conditioning for controllable sampling
- Ran qualitative and compute tradeoff studies across step counts and guidance scales; curated result grids and failure cases for prompt sensitivity and artifacts

Technical Skills

Languages: Python, C++, Java, SQL, TypeScript, JavaScript, HTML, CSS
Tools: Git, Linux, Docker, AWS, PostgreSQL, DuckDB, GitHub Actions (CI/CD)
Libraries / Frameworks: PyTorch, OpenCV, NumPy, pandas, scikit-learn, FastAPI, Next.js (React)